

Criteria-Augmented Re-scoring of the Hide-One-Diagnosis Benchmark

100-patient MIMIC-IV pilot, Haiku predictor

Thousand Health — Internal Report

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TL;DR

On a 100-patient MIMIC-IV hide-one-diagnosis pilot using Haiku as the predictor, strict ICD-label matching scores the system at **34% top-3 sensitivity**. After re-scoring with a 522-entry textbook criteria library (verbatim quotes from Tier 1–3 sources, re-verified against cited URLs by LLM verifier agents; *not* physician-reviewed) and an LLM judge that checks each chart against verbatim guideline thresholds, the **criterion-validated sensitivity rises to 78%** (+44 patients), at a candidate-level **lenient PPV of 66%** (strict 41%). The criteria layer is doing the real precision work; Haiku’s confidence score adds little additional separation.

1 Background

The hide-one-diagnosis benchmark hides a single ICD-coded diagnosis from the chart and asks the LLM to recover it as one of its top-3 ranked candidates. Strict scoring requires an exact string/concept match against the hidden ICD label. This scoring under-counts cases where the LLM names a different but clinically real condition the patient has — e.g., the hidden ICD is “Sialolithiasis” but the LLM correctly identifies “Hypokalemia” that the chart in fact documents.

We hypothesized that many of these “misses” are not errors but legitimate co-existing conditions the strict scorer cannot validate. To test this, we built `criteria.db.json` — 522 conditions with verbatim quoted textbook criteria from Tier 1–3 sources (specialty societies, government regulatory, peer-reviewed foundational papers). Source-verbatim fidelity was checked by LLM verifier agents that re-fetched each cited URL and confirmed the quoted text appears at the source. No physician reviewer has audited the library.

2 Method

2.1 Inputs

- **Predictor:** Claude Haiku, top-3 ranked candidates with a 0–100 confidence score per candidate, across 100 MIMIC-IV admissions.
- **Criteria DB:** 522 conditions, each with verbatim `criterion_quote` and source URL. 6 entries dropped during the trust audit for citation errors; 76 carry caveat notes from the audit.

- **Charts:** De-identified MIMIC-IV chart.txt per admission (PHI/DUA-protected; not committed).

2.2 Pipeline

1. **Slug mapping.** LLM mapper assigned each of 218 unique top-3 candidate dx strings to a criteria_db slug or null. 157 mapped (72%); 61 unmappable (vague compound dx, unsupported conditions).
2. **Criterion-check judging.** For each of 230 (patient, candidate) pairs that mapped, a Sonnet judge agent read `chart.txt`, applied the verbatim criterion quote strictly, and returned `meets` / `does_not_meet` / `insufficient_evidence`. Ten parallel agents, ~23 jobs each.
3. **Re-aggregation.** A candidate is a true positive if it either matches the strict hidden ICD label or the judge said `meets`. A patient is a true positive if any of its top-3 candidates is a TP.

2.3 Metric definitions

Sensitivity (per-patient) $TP_{patient}/100$, where a patient is a TP if ≥ 1 top-3 candidate above the confidence threshold is validated. Per-patient is the only well-defined unit for sensitivity here — the universe of “all conditions the LLM should have produced” is unbounded once we leave the strict hidden-ICD scope.

PPV strict (per-candidate) $TP/allcandidatesabovethreshold$. Counts `does_not_meet`, `insufficient_evidence` and unmappable candidates all as false positives. Harsh on the predictor: it pays for our redacted labs and incomplete criteria DB.

PPV lenient (per-candidate) $TP/(TP + does_not_meetonly)$. Drops unknowns from the denominator; only counts candidates where the judge actively saw chart evidence contradicting the criterion as FP. Fair to the predictor.

3 Results

3.1 Aggregate verdict mix

Across 230 graded (patient, candidate) pairs:

Verdict	Count
meets	103
does_not_meet	65
insufficient_evidence	62

Of the 297 total top-3 candidates: 230 were judgable (slug-mapped), 49 had no slug, 18 had no verdict for other reasons (slug existed but judge had no input row).

3.2 Patient-level TP rate

Metric	Patients	Rate
Strict top-3 (hidden ICD recovered)	34/100	34%
Criterion-met top-3 (potential TP only)	67/100	67%
Union (strict OR criterion-met)	78/100	78%
Net new TPs from criteria layer	+44	

Cohort breakdown:

Original-scoring cohort	<i>n</i>	Criterion-met in top-3
Strict TP (hidden at rank 1–3)	34	23 (sanity check)
Matched rank 4–10 (off-podium)	24	16 (66%)
No match in top-10 (rank –1)	42	28 (66%)

The 23/34 “strict TP also passes criteria” sanity check is reasonable; the other 11 strict TPs had hidden ICDs that did not have a criteria_db entry, so the criterion machinery had nothing to validate.

3.3 Confidence-threshold sweep

conf $\geq$	#cand	TP	PPV strict	PPV lenient	Sens relaxed	Sens strict
0	297	121	41%	66%	78%	34%
60	295	121	41%	67%	78%	34%
65	290	120	41%	67%	77%	34%
70	243	104	43%	68%	70%	31%
75	162	76	47%	75%	56%	24%
80	69	36	52%	78%	25%	15%
85	43	26	60%	84%	21%	13%
90	14	10	71%	91%	10%	7%

3.4 Confidence calibration

Mean confidence by ground-truth category:

Category	Mean confidence
Strict hidden-match candidates ($n = 30$)	80.7
“meets” candidates ($n = 103$)	77.7
“does_not_meet” candidates ($n = 65$)	73.6

A 7-point spread between correct and incorrect predictions. Haiku’s confidence is only weakly calibrated — it is not a reliable filter on its own.

4 Interpretation

4.1 What the +44 actually represents

The 44 new TPs are cases where Haiku named a different real condition the patient has — validated against chart data using verbatim textbook thresholds. Examples from the “no-match” cohort

($n = 28$ new TPs):

- Hidden: *Sialolithiasis* → Haiku ranked Hypokalemia #1; K was in fact low.
- Hidden: *Pressure ulcer, stage III* → Hypokalemia + Hypophosphatemia, both verified.
- Hidden: *Melena* → Hypomagnesemia + Anemia, both verified.
- Hidden: *Cervicalgia* → Anemia (normocytic) + Hypomagnesemia, both verified.

This pattern is loud in the data: “no-match” admissions frequently have a relatively benign hidden ICD (cervicalgia, glaucoma, urticaria), while Haiku correctly surfaces the clinically significant electrolyte / AKI / anemia / sepsis picture — which a physician reading the chart would also prioritize.

4.2 Where PPV lives

- No threshold: PPV strict 41%, PPV lenient 66%, Sens 78%.
- $\text{conf} \geq 75$: PPV strict 47%, PPV lenient 75%, Sens 56%.
- $\text{conf} \geq 80$: PPV strict 52%, PPV lenient 78%, Sens 25%.

The 75 cutoff is the best balanced operating point under Haiku — modest precision gain, modest recall cost. Above 80 the sensitivity collapse is non-linear because Haiku’s confidence distribution is compressed ($p_{10} = 68$, $p_{90} = 88$).

4.3 Caveats

1. **Electrolyte / AKI dominance.** ~70% of the “meets” verdicts are single-lab-threshold conditions (hypoK, hypoMg, hypoCa, hypoP, AKI by KDIGO). These are easy to verify and over-represented in top-3 candidates. “Real test-of-skill” specific syndromes are a much smaller share.
2. **62 insufficient_evidence.** The chart often had redacted lab values (---). Strict PPV penalizes the predictor for our redactions; the gap between strict and lenient PPV is mostly this.
3. **Slug-mapper was conservative.** 61/218 unique dx strings unmapped. Some truly are unmappable (multi-condition free text); some are criteria-DB coverage gaps.
4. **Predictor is Haiku.** Confidence is poorly calibrated. Re-running on Sonnet/Opus would likely shift the curve.
5. **No physician in the loop, anywhere.** The criteria DB was source-verified by LLM agents, not reviewed by clinicians. The 230 chart-vs-criterion verdicts were rendered by a single LLM judge (Sonnet), with no physician adjudication. Both layers can carry systematic errors a domain expert would catch.

5 Implications for product framing

- As an **autonomous flagger** (no physician in loop): 41% top-3 strict PPV is too noisy. A $\text{conf} \geq 90$ slice yields 91% lenient PPV but 10% sensitivity — a niche “high-confidence-only auto-alert” could be viable.
- As a **triage assistant** (physician reviews flagged cases): 78% relaxed sensitivity at 66% lenient PPV is a defensible operating point. The criteria layer transforms the system from a string-match benchmark into a clinically-grounded detector.
- **The criteria DB is the asset.** Without it, sensitivity is 34%; with it, 78%. Confidence-thresholding adds little independent lift. The pitch should center on verbatim society-guideline criteria as the verification layer, not the LLM’s self-reported confidence.

6 Next steps

1. Re-run with Sonnet/Opus as predictor; quantify confidence-calibration delta.
2. Expand criteria_db coverage for unmappable dx strings (61 unique).
3. Spot-check 20–30 “meets” verdicts manually for judge-error rate.
4. Replace MDCalc / Wikipedia / weak-domain sources in the 76 caveat entries with primary society guideline citations.

Artifacts. Re-score detail: `/tmp/rescore_result.json` (local, not committed — contains PHI-derived data). Judge verdicts: `/tmp/verdicts_judge_0..9.json`. Criteria DB: `criteria_db.json` (522 entries post-audit).